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Dilute nitride long-wavelength emitter with improved performance over temperature

Abstract

In some implementations, a method may include forming a quantum well (QW) layer using an epitaxial growth process, where the epitaxial growth process is performed according to a first growth mode to form the QW layer. The method may include forming a quantum well barrier (QWB) layer using the epitaxial growth process, where the epitaxial growth process is performed according to a second growth mode to form the QWB layer. In some implementations, a nitrogen flux used in the first growth mode is different from a nitrogen flux used in the second growth mode. In some implementations, a gallium flux used in the first growth mode is different from a gallium flux used in the second growth mode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This patent application claims priority to U.S. Provisional Patent Application No. 63/268,099, filed on Feb. 16, 2022, and entitled “DILUTE NITRIDE LONG-WAVELENGTH (>1200 NM) VCSELS WITH IMPROVED PERFORMANCE OVER TEMPERATURE.” The disclosure of the prior application is considered part of and is incorporated by reference into this patent application.

TECHNICAL FIELD

(1) The present disclosure relates generally to a vertical-cavity surface-emitting laser (VCSEL) and, more particularly, to a dilute nitride long-wavelength VCSEL with an active region that

improves performance of the VCSEL over temperature.

BACKGROUND

(2) A dilute nitride material, such as indium gallium arsenide nitride (InGaAsN) or InGaAsN antimony (InGaAsNSb), can be used as a material for an active region of a GaAs-based emitter (e.g., a VCSEL, an edge-emitting laser, or the like) to enable lasing at a long wavelength (e.g., a wavelength in a range from approximately 1200 nanometers (nm) to approximately 1600 nm).

SUMMARY

(3) In some implementations, a method includes forming a quantum well (QW) layer using an epitaxial growth process, the epitaxial growth process being performed according to a first growth mode to form the QW layer; and forming a quantum well barrier (QWB) layer using the epitaxial growth process, the epitaxial growth process being performed according to a second growth mode to form the QWB layer, wherein at least one of: a nitrogen flux used in the first growth mode is different from a nitrogen flux used in the second growth mode, or a gallium flux used in the first growth mode is different from a gallium flux used in the second growth mode.

(4) In some implementations, a dilute nitride long-wavelength emitter includes a cladding layer; an active region including one or more QW layers and one or more QWB layers; and a barrier layer between the active region and the cladding layer, wherein the barrier layer comprises a graded-index structure.

(5) In some implementations, a dilute nitride long-wavelength emitter includes an active region including: one or more QW layers; and one or more QWB layers, wherein at least one QWB of the one or more QWB layers includes a p-doped region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of an example of light-current-voltage (L-I-V) characteristics of a conventional dilute nitride long-wavelength VCSEL at different temperatures.

(2) FIGS. 2A-2C are diagrams associated with an example implementation of a dilute nitride long-wavelength emitter array with improved performance over temperature, as described herein.

(3) FIGS. 3A-3D illustrate simplified band diagrams of example implementations of an a cavity and active region of a dilute nitride long-wavelength emitter with improved performance over temperature, as described herein.

(4) FIG. 4 is a flowchart of an example process associated with forming a dilute nitride long-wavelength emitter with improved performance over temperature, as described herein.

DETAILED DESCRIPTION

(5) The following detailed description of example implementations refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

(6) A dilute nitride material (e.g., InGaAsN, InGaAsNSb, or the like) can be used as a material for an active region of a GaAs-based emitter to enable long-wavelength lasing (e.g., lasing at a wavelength in a range from approximately 1200 nm to approximately 1600 nm). However, dilute nitride materials have relatively poor carrier confinement in a conventional quantum well (QW) active region, such as an active region that includes a typical InGaAsNSb/GaAsN heterostructure. The poor carrier confinement in the QW active region causes performance of the emitter to degrade significantly as temperature of the emitter increases. FIG. 1 is a diagram illustrating an example of light-current-voltage (L-I-V) characteristics of a conventional dilute nitride long-wavelength VCSEL at different temperatures. In the example shown in FIG. 1, the VCSEL is lasing at approximately 1380 nm and comprises an InGaAsNSb/GaAsN active region grown on a GaAs substrate. As illustrated in FIG. 1, at a comparatively higher operating temperature (e.g., 50 degrees

Celsius ($^{\circ}$ C.) as compared to 25° C.), output power of the conventional long-wavelength VCSEL at a given voltage and current is degraded, meaning that performance of the conventional long-wavelength VCSEL degrades as temperature increases.

(7) Some implementations described herein provide a dilute nitride long-wavelength emitter (e.g., a dilute nitride long-wavelength VCSEL) with improved performance over temperature. In some implementations, improved performance of the dilute nitride long-wavelength emitter over temperature is achieved by improving carrier confinement, improving carrier injection, or improving capture efficiency of the dilute nitride long-wavelength emitter. In some implementations, improved carrier confinement, improved carrier injection, or improved capture efficiency can be achieved by increasing a quantum well barrier (QWB) band gap in the dilute nitride long-wavelength emitter, by including a graded-index outer barrier in the dilute nitride long-wavelength emitter, or by including a p-doped region in a QWB of the dilute nitride long-wavelength emitter. Additional details are described below.

(8) FIGS. 2A-2C are diagrams associated with an example implementation of a dilute nitride long-wavelength emitter array **200** with improved performance over temperature. FIG. 2A is a diagram illustrating various layers of the emitter array **200**. In some implementations, the emitter array **200** includes a single emitter. In some implementations, the emitter array **200** includes an array of emitters. In some implementations, the emitters of the emitter array **200** include one or more top-emitting emitters or one or more bottom-emitting emitters. As shown in FIG. 2A, the emitter array **200** may include a substrate **202**, a bottom metal **204**, a bottom mirror structure **206**, a cavity with dilute nitride active region **208** (herein referred to as cavity/active region **208**), an optical aperture (OA) layer **210**, a top mirror structure **212**, a top contact layer **214**, and a top metal **216**.

(9) Substrate **202** includes a supporting material upon which or within which one or more layers or features of the emitter array **200** are grown or fabricated. In some implementations, the substrate **202** comprises an n-type material. In some implementations, the substrate **202** comprises a semi-insulating type of material. In some implementations, the semi-insulating type of material may be used when the emitter array **200** includes one or more bottom-emitting emitters in order to reduce optical absorption from the substrate **202**. In such an implementation, the emitter array **200** may include a contact buffer in or near the bottom mirror structure **206**. In some implementations, the substrate **202** may be formed from a semiconductor material, such as gallium arsenide (GaAs), indium phosphide (InP), or another type of semiconductor material. In some implementations, a bottom contact (e.g., a bottom n-contact) of the emitter array **200** can be made from a backside of the substrate **202**. In some implementations, the bottom contact of the emitter array **200** can be made from a front side of the emitter array **200**. In some implementations, the front side contact can be achieved by, for example, etching a mesa step or trench to the substrate **202**, or inserting a contact buffer in or near the bottom mirror structure **206**.

(10) Bottom metal **204** includes a metal layer on a bottom surface of the substrate **202** (e.g., at a backside of the emitter array **200**). In some implementations, the bottom metal **204** is formed from an n-type material. In some implementations, the bottom metal **204** is a layer that makes electrical contact with the substrate **202**. In some implementations, the bottom metal **204** serves as a cathode for the emitter array **200**. In some implementations, the bottom metal **204** may serve as an anode for the emitter array **200**. For example, in some implementations, the bottom metal **204** may serve as a common anode for a group of sub-arrays of a VCSEL array, where the emitter array **200** is one of the group of sub-arrays. In some implementations, the bottom metal **204** may include an annealed metallization layer, such as a gold-germanium-nickel (AuGeNi) layer, a palladium-germanium-gold (PdGeAu) layer, among other examples.

(11) Bottom mirror structure **206** is a bottom reflector of an optical resonator of the emitter array **200**. For example, the bottom mirror structure **206** may include a distributed Bragg reflector (DBR), a dielectric mirror, or another type of mirror structure. In some implementations, the bottom mirror structure **206** is formed from an n-type material. In some implementations, the

bottom mirror structure **206** is on a top surface of the substrate **202**. In some implementations, the bottom mirror structure **206** may have a thickness in a range from approximately 3.5 microns (μm) to approximately 9 μm , such as 5 μm . In some implementations, the bottom mirror structure **206** includes a set of layers (e.g., aluminum gallium arsenide (AlGaAs) layers) grown using a metal-organic chemical vapor deposition (MOCVD) technique, a molecular beam epitaxy (MBE) technique, or another technique.

(12) Cavity/active region **208** includes one or more layers where electrons and holes recombine to emit light and define the emission wavelength range of the emitter array **200**, where one or more QWs of the cavity/active region **208** are formed from a dilute nitride material. The dilute nitride material may in some implementations comprise a composite semiconductor using a combination of InGaAlAsPSb with a low percentage (e.g., less than approximately 5%) of nitrogen added to a Group V site to remain lattice-matched or lattice constant closer to a GaAs substrate to mitigate forming dislocations. The dilute nitride material from which the one or more QWs of the cavity/active region **208** are formed may include, for example, dilute nitride InGaAsN or InGaAsNSb. In some implementations, the cavity/active region **208** may include one or more cavity spacer layers. In some implementations, the one or more cavity spacer layers may enable epitaxial growth to have sufficient room for ramping compositions or temperature. In some implementations, the one or more cavity spacer layers may reduce strain between active regions of the cavity/active region **208**. In some implementations, the one or more cavity spacer layers may mitigate thermal issues of laser operation. In some implementations, the one or more cavity spacer layers may include an oxidation layer. The optical thickness of the cavity/active region **208** (including cavity spacer layers), the top mirror structure **212**, and the bottom mirror structure **206** define the resonant cavity wavelength of the emitter array **200**, which may be designed within an emission wavelength range of the cavity/active region **208** to enable lasing. In some implementations, the cavity/active region **208** may be formed on the bottom mirror structure **206**. In some implementations, the cavity/active region **208** may have a thickness in a range from approximately 0.006 μm to approximately 0.5 μm , such as 0.15 μm or 0.30 μm . In some implementations, the cavity/active region **208** includes a set of layers grown using an MOCVD technique, an MBE technique, or another technique.

(13) In some implementations, the cavity/active region **208** may be designed so as to improve performance of the emitter array **200** over temperature, as described herein. Additional details regarding the cavity/active region **208** are provided below with respect to FIGS. 2A-2C and FIGS. 3A-3D.

(14) OA layer **210** is a layer for forming an aperture that provides optical and electrical confinement for the VCSELs of the emitter array **200**. In some implementations, the OA layer **210** enhances carrier and mode confinement of the emitter array **200** and, therefore, can improve performance of the emitter array **200**. In some implementations, the OA layer **210** is on, under, or in the cavity/active region **208**. In some implementations, there may be one or more spacer layers or mirror layers (e.g., DBRs) between the OA layer **210** and the cavity/active region **208**. In some implementations, the OA layer **210** is on a side of the cavity/active region **208** nearer to the bottom mirror structure **206** (i.e., on a substrate side of the cavity/active region **208**). In some implementations, the OA layer **210** is on a side of the cavity/active region **208** nearer to the top mirror structure **212** (i.e., on a non-substrate side of the cavity/active region **208**). In some implementations, the emitter array **200** may include one or more OA layers **210**.

(15) In some implementations, the OA layer **210** is an oxide layer formed as a result of oxidation of one or more epitaxial layers of the emitter array **200**. For example, the OA layer **210** may be an aluminum oxide (Al₂O₃) layer formed as a result of oxidation of an epitaxial layer (e.g., an AlGaAs layer, an AlAs layer, and/or the like). In some implementations, the OA layer **210** may have a thickness in a range from approximately 0.007 μm to approximately 0.04 μm , such as 0.02 μm . In some implementations, oxidation trenches (not shown in FIG. 2A, and shown as filled in

FIG. 2B) etched around the VCSELs in the emitter array **200** may allow steam to access the epitaxial layer(s) from which the OA layer **210** is formed. In some implementations, the optical aperture has a circular shape. In some implementations, the optical aperture has a non-circular shape. In some implementations, a size (e.g., a diameter) of the optical aperture formed by the OA layer **210** is in a range from approximately 1 μm to approximately 300 μm , such as 5 μm or 8 μm . In some implementations, in addition to the one or more OA layers **210**, the emitter array **200** may include one or more other types of structures or layers that provide current confinement, such as an implant passivation structure, a mesa isolation structure, a moat trench isolation structure, a buried tunnel junction, or the like. Additionally, or alternatively, such other types of structures or layers for providing current confinement may in some implementations be included in or integrated with one or more OA layers (e.g., one or more OA layers **210**) of the emitter array **200**.

(16) Top mirror structure **212** is a top reflector of the optical resonator of the emitter array **200**. For example, the top mirror structure **212** may include a DBR, a dielectric mirror, and/or the like. In some implementations, the top mirror structure **212** is formed from a p-type material. In some implementations, the top mirror structure **212** may have a thickness in a range from approximately 1 μm to approximately 6 μm , such as 3 μm . In some implementations, the top mirror structure **212** includes a set of layers (e.g., AlGaAs layers) grown using an MOCVD technique, an MBE technique, or another technique. In some implementations, the top mirror structure **212** is grown on or over the cavity/active region **208**. In some implementations, a total thickness from a bottom surface of the bottom mirror structure **206** to a top surface of the top mirror structure **212** may be in a range from, for example, approximately 4.5 μm to approximately 26.4 μm , such as approximately 8.6 μm . In some implementations, a thickness of one or more of the layers of the emitter array **200** may be selected in order to provide a structure that achieves high reflectivity (e.g., greater than approximately 99% reflectivity). In some implementations, a smaller total thickness may facilitate growth time reduction of the emitter array **200** or stress reduction within the emitter array **200**.

(17) The top contact layer **214** is a top contact layer of the emitter array **200** that makes electrical contact with the top mirror structure **212** through which current may flow. In some implementations, the top contact layer **214** includes an annealed metallization layer. For example, the top contact layer **214** may include a chromium-gold (Cr—Au) layer, a gold-zinc (Au—Zn) layer, a titanium-platinum-gold (TiPtAu) layer, a gold-germanium-nickel (AuGeNi) layer, a palladium-germanium-gold (PdGeAu) layer, or the like. In some implementations, the top contact layer **214** has a thickness in a range from approximately 0.03 μm to approximately 0.3 μm , such as 0.2 μm . In some implementations, the top contact layer **214** has a ring shape, a slotted ring shape, a tooth wheel shape, or another type of circular or non-circular shape (e.g., depending on a design of the VCSELs in the emitter array **200**).

(18) The top metal **216** is a top metal layer at a front side of the emitter array **200**. In some implementations, the top metal **216** is formed from a p-type material. Alternatively, in some implementations, the top metal **216** is formed from an n-type material. In some implementations, the top metal **216** may be a layer that makes electrical contact with the top contact layer **214**. In some implementations, the top metal **216** serves as an anode for the emitter array **200**. In some implementations, the top metal **216** may serve as a cathode for the emitter array **200**. For example, in some implementations, the top metal **216** may serve as an isolated cathode for a particular sub-array of a VCSEL array including a group of sub-arrays, where the emitter array **200** is one of the group of sub-arrays.

(19) FIG. 2B is a diagram illustrating an example of a cross section of a top-emitting VCSEL in the emitter array **200** with the layers shown in FIG. 2A. As shown in FIG. 2B, the top metal **216** is insulated from sidewalls of the trenches by a dielectric layer **218**, which may be, for example, silicon nitride (SiN), silicon dioxide (SiO₂), a polymer dielectric, or another type of insulating material. Additionally, as shown in FIG. 2B, the VCSEL may have an isolation implant **220** to prevent free carriers from reaching edges of the trenches and/or to isolate adjacent VCSELs in the

emitter array **200** from one another (e.g., if the trenches do not fully enclose the VCSELs of the emitter array **200**).

(20) FIG. 2C is a diagram illustrating an example of a cross section of a bottom-emitting VCSEL in the emitter array **200** with the layers shown in FIG. 2A. The structure of the bottom-emitting VCSEL may be similar to that of the top-emitting VCSEL shown in FIG. 2B with some exceptions. For example, as illustrated in FIG. 2C, in the case of the bottom-emitting VCSEL, an opening is present in the bottom metal **204** to permit emission of light out of the substrate **202**. In some implementations, as shown in FIG. 2C, an anti-reflective (AR) coating **222** may be formed in the opening in the bottom metal **204**. As further shown in FIG. 2C, in the bottom-emitting VCSEL, the top metal **216** covers the top surface of the VCSEL emitter over the top contact layer **214**. Other examples of structural differences between the top-emitting VCSEL and the bottom-emitting VCSEL may include the top contact layer **214** and the top metal **216** being differently formed in the bottom-emitting VCSEL, the positions of the top mirror structure **212** and the bottom mirror structure **206** being reversed (not illustrated in FIG. 2C) in the bottom-emitting VCSEL, the substrate **202** being thinned in the bottom-emitting VCSEL (not illustrated in FIG. 2C), or one or more OA layers (e.g., one or more OA layers **210**) being differently formed, among other examples.

(21) The number, arrangement, thicknesses, order, symmetry, or the like, of layers shown in FIGS. 2A-2C are provided as examples. In practice, emitter array **200** may include additional layers, fewer layers, different layers, differently constructed layers, or differently arranged layers than those shown in FIGS. 2A-2C. For example, the emitter array **200** may include one or more cladding layers, one or more spacer layers (e.g., near the bottom mirror structure **206**, the OA layer **210**, and/or the top mirror structure **212**), one or more additional mirror structures, or the like, not shown in FIGS. 2A-2C. Additionally, or alternatively, a set of layers (e.g., one or more layers) of emitter array **200** may perform one or more functions described as being performed by another set of layers of emitter array **200**, and any layer may comprise more than one layer.

(22) FIGS. 3A-3D illustrate simplified band diagrams of example implementations of the cavity/active region **208** of the emitter array **200**, as described herein.

(23) Conventionally, when using plasma-assisted MBE to grow a dilute nitride active region (e.g., an active region including an InGaAsNSb QW and a GaAsN QWB), gallium flux and atomic nitrogen flux are kept constant to mitigate potential growth variation. However, maintaining a constant gallium flux and atomic nitrogen flux provides a band gap between a given QW and an adjacent QWB (herein referred to as a QWB band gap) that is insufficient to achieve carrier confinement that enables adequate performance at higher temperatures. Therefore, to enable adequate performance at higher temperatures, the cavity/active region **208** may in some implementations have an increased QWB band gap (e.g., as compared to a QWB band gap in a conventional dilute nitride long-wavelength emitter).

(24) FIG. 3A is a simplified band diagram of an example implementation of the cavity/active region **208** in which improved carrier confinement, carrier injection, and/or capture efficiency is achieved by increasing a QWB band gap. As indicated in the band diagram shown in FIG. 3A, the cavity/active region **208** may include a p-doped cladding layer **302**, a p-side non-intentionally doped (pNID) layer **303**, a group of QWB layers **304**, a group of QW layers **306**, an n-side non-intentionally doped (nNID) layer **308**, and an n-doped cladding layer **310**.

(25) The cladding layers of the cavity/active region **208**—the p-doped cladding layer **302** and the n-doped cladding layer **310**—are layers associated with injecting charge carriers and providing light confinement in the emitter array **200**. For example, when a forward bias is applied, electrons are injected from the n-doped cladding layer **310** and into the active layers of the cavity/active region **208** (e.g., into the QW layers **306** and the QWB layers **304**), and holes are supplied from the p-doped cladding layer **302**. As indicated in FIG. 3A, the p-doped cladding layer **302** and the n-doped cladding layer **310** have a higher band gap energy than the QW layers **306** and the QWB layers **304**. Therefore, the p-doped cladding layer **302** and the n-doped cladding layer **310** prevent

the injected electrons and holes from diffusing across the emitter junction. Here, the electrons and holes are confined to the active layers of the cavity/active region **208** to create a state of population inversion, thereby allowing amplification of light by stimulated emission. In some implementations, the p-doped cladding layer **302** may be, for example, a carbon-doped GaAs layer with a doping level in a range from approximately $1 \times 10^{16} \text{ cm}^{-3}$ to approximately $4 \times 10^{18} \text{ cm}^{-3}$, such as $1 \times 10^{17} \text{ cm}^{-3}$. In some implementations, the p-doped cladding layer **302** may have a thickness in a range from approximately 50 μm to approximately 2000 μm , such as 200 μm . In some implementations, the n-doped cladding layer **310** may be, for example, a silicon-doped GaAs layer with a doping level in a range from approximately $1 \times 10^{16} \text{ cm}^{-3}$ to approximately $4 \times 10^{18} \text{ cm}^{-3}$, such as $5 \times 10^{17} \text{ cm}^{-3}$. In some implementations, the n-doped cladding layer **310** may have a thickness in a range from approximately 50 μm to approximately 2000 μm , such as 200 μm .

(26) In some implementations, the pNID layer **303** is formed between the p-doped cladding layer **302** and a QWB layer **304** nearest to the p-doped cladding layer **302**. In some implementations, the pNID layer **303** serves to ensure that p-type doping is not applied to other layers of the cavity/active region **208**, such as the QWB layer **304** nearest to the p-doped cladding layer **302**. As indicated in FIG. 3A, pNID layer **303** may have a band gap energy that matches that of the p-doped cladding layer **302**. In some implementations, the pNID layer **303** may be, for example, a GaAs layer with a thickness in a range from approximately 0 nm to approximately 500 nm, such as 5 nm.

(27) In some implementations, the nNID layer **308** is formed between the n-doped cladding layer **310** and a QWB layer **304** nearest to the n-doped cladding layer **310**. In some implementations, the nNID layer **308** serves to ensure that n-type doping is not applied to other layers of the cavity/active region **208**, such as the QWB layer **304** nearest to the n-doped cladding layer **310**. As indicated in FIG. 3A, nNID layer **308** may have a band gap energy that matches that of the n-doped cladding layer **310**. In some implementations, the nNID layer **308** may be, for example, a GaAs layer with a thickness in a range from approximately 0 nm to approximately 500 nm, such as 50 nm.

(28) The QW layers **306** and the QWB layers **304** of the cavity/active region **208** form the active layers (also referred to as a gain region or gain medium) in which lasing of the emitter array **200** is achieved. As shown in FIG. 3A, the cavity/active region **208** may in some implementations include one or more QW layers **306**, and each QW layer **306** may be between a pair of QWB layers **304**. As shown in FIG. 3A, an energy band gap of a given QWB layer **304** is higher than that of a given QW layer **306**. As further shown, an energy band gap of a given QWB layer **304** is lower than that of the p-doped cladding layer **302**, the pNID layer **303**, the nNID layer **308**, and the n-doped cladding layer **310**. In some implementations, a QW layer **306** may be, for example, an InGaAsNSb layer with a thickness in a range from approximately 5 nm to approximately 12 nm, such as 7 nm. In some implementations, a QWB layer **304** may be, for example, a GaAsN layer with a thickness in a range from 5 nm to approximately 40 nm, such as 20 nm.

(29) As noted above, the band gap of the QWB layers **304** of the cavity/active region **208** may in some implementations be increased (e.g., as compared to a conventional dilute nitride long-wavelength emitter) to improve performance of the emitter array **200** over temperature. In some implementations, to increase the band gap of the QWB layer **304**, a percentage of nitrogen (N %) can be reduced (e.g., by approximately 20% or more relative to a conventional QWB). In some implementations, the percentage of nitrogen can be reduced by decreasing a nitrogen flux or by increasing a growth rate during growth of the cavity/active region **208**. In some implementations, the nitrogen flux can be adjusted by changing a plasma source setting used during growth, such as a power setting or a nitrogen (N_{2}) gas flow setting. For comparison, it may be desirable to adjust gallium flux by switching between, or combining the use of, multiple gallium sources to increase the growth rate, which reduces the percentage of nitrogen in a given QWB layer **304**.

(30) In some implementations, in association with forming the cavity/active region **208**, a QW

layer **306** may be formed using an epitaxial growth process. During growth of the QW layer **306**, the epitaxial growth process may be performed according to a first growth mode. Next, a QWB layer **304** may be formed using the epitaxial growth process. However, during growth of the QWB layer **304**, the epitaxial growth process may be performed according to a second growth mode. Here, a nitrogen flux used in the first growth mode may be different from a nitrogen flux used in the second growth mode, or a gallium flux used in the first growth mode may be different from a gallium flux used in the second growth mode. In this way, the different growth modes may be used in association with forming the QW layers **306** and the QWB layers **304** of the cavity/active region **208**, thereby enabling formation of QWB layers **304** with higher band gaps (e.g., as compared to the QWB band gap in a conventional dilute nitride long-wavelength emitter).

(31) The dotted line in FIG. 3A indicates the band gap structure of a conventional dilute nitride long-wavelength emitter. As shown by comparing the dotted line in FIG. 3A and the upper solid line in FIG. 3A, the QWB layers **304** of the cavity/active region **208** have a higher band gap than the QWB band gap in the conventional dilute nitride long-wavelength emitter. As a result, carrier confinement, carrier injection, and/or capture efficiency can be improved.

(32) As further shown in FIG. 3A, the cavity/active region **208** may in some implementations be formed to include an interlayer (IL) **312**. As shown in FIG. 3A, the interlayer **312** may have a band gap that is lower than that of a given QWB layer **304** and that is higher than that of a given QW layer **306**. That is, the interlayer **312** may have a band gap that is between that of a QW layer **306** and a QWB layer **304**. In some implementations, the interlayer **312** improves carrier injection into a given QW layer **306** (e.g., by providing a band gap “transition” between the higher band gap of the QWB layer **304** and the lower band gap of the QW layer **306**). In some implementations, the interlayer **312** may reduce a growth interference risk (e.g., when using a different gallium source or changing a nitrogen plasma source setting to accommodate an optimal growth condition for a QW layer **306**). In some implementations, the interlayer **312** may be, for example, a GaAsN layer that has a thickness in a range from approximately 0.1 nm to 10.0 nm, such as 1 nm. In some implementations, the interlayer **312** may be formed using the epitaxial growth process, and during growth of the interlayer **312** the epitaxial growth process may be performed according to the first growth mode (e.g., the growth mode used in association with growing a given QW layer **306** of the cavity/active region **208**).

(33) In some implementations, improved carrier confinement, improved carrier injection, or improved capture efficiency can be achieved by including a graded-index structure in the cavity/active region **208**. FIGS. 3B and 3C are simplified energy band diagrams of example implementations of the cavity/active region **208** in which improved carrier confinement, carrier injection, or capture efficiency is achieved by including a graded-index structure **314** in the cavity/active region **208**. In some implementations, a percentage of nitrogen in the graded-index structure increases in the graded-index structure **314** in a direction moving toward a center of the cavity/active region **208** (e.g., a direction from the pNID layer **303** toward a nearest QWB layer **304** layer, a direction from the nNID layer **308** toward a nearest QWB layer **304**). In some implementations, the graded-index structure **314** may be, for example, a graded-index separate confinement heterostructure (GRINSCH).

(34) In some implementations, as illustrated in FIGS. 3B and 3C, the graded-index structure **314** may be formed as an outer barrier layer of the cavity/active region **208** (e.g., a layer between the pNID layer **303** and a nearest QWB layer **304**, a layer between the nNID layer **308** and a nearest QWB layer **304**). Additionally, or alternatively, a graded-index structure **314** may be formed in the pNID layer **303** and/or the p-doped cladding layer **302**. Additionally, or alternatively, a graded-index structure **314** may be formed in the nNID layer **308** and/or the n-doped cladding layer **310**. Additionally, or alternatively, a graded-index structure **314** may be partially doped. For example, a graded-index structure **314** may be doped in layers nearer to the p-doped cladding layer **302**, or the n-doped cladding layer **310**, in the absence of the pNID layer **303**, nNID layer **308**, respectively. In

some implementations, a graded-index structure may be a GaAsN layer (in which a percentage of nitrogen varies) having a thickness in a range from approximately 5 nm to approximately 500 nm, such as 20 nm.

(35) In some implementations, the graded-index structure **314** reduces carrier leakage and enhances carrier injection and capture into the QW layers **306** of the cavity/active region **208**. In some implementations, the graded-index structure **314** may have a linear grading, an example of which is shown in FIG. 3B. Alternatively, in some implementations, the graded-index structure **314** may have a step grading (i.e., a series of step-grades), an example of which is shown in FIG. 3C. In some implementations, the graded-index structure may have a combination of step-grades and linear grades. In some implementations, the type of grading may be selected based on, for example, which growth mode is simpler or more reliable in dilute nitride epitaxial growth.

(36) In some implementations, improved carrier confinement, improved carrier injection, or improved capture efficiency can be achieved by including a p-doped region in one or more QWB layers **304** of the cavity/active region **208**. FIG. 3D is a simplified energy band diagram of an example implementation of the cavity/active region **208** in which improved carrier confinement, carrier injection, or capture efficiency is achieved by including one or more p-doped regions **316** in one or more QWB layers **304** of the cavity/active region **208**.

(37) In some implementations, a p-doped region **316** may be a carbon p-type doped region. In some implementations, a doping level of the p-type doping of a given p-doped region **316** may be in a range from approximately $1 \times 10^{16} \text{ cm}^{-3}$ to approximately $1 \times 10^{19} \text{ cm}^{-3}$, such as $5 \times 10^{17} \text{ cm}^{-3}$. In some implementations, a p-doped region **316** can be formed at or near a center region of a QWB layer **304**. In some implementations, the p-doped region **316** has a thickness in a range from approximately 0.2 nm to approximately 20 nm, such as 2 nm. In some implementations, the p-doped region **316** has a thickness that matches a thickness of the QWB layer **304**. In some implementations, delta-doping may be utilized to form a given p-doped region **316**.

(38) Notably, p-type doping in a QWB layer **304** could in some cases degrade photoluminescence (PL) intensity or cause wavelength shift. Further, p-type doping can increase cavity absorption. Therefore, in some implementations, the doping level of a given p-doped region **316** and doped layer thickness of the given p-doped region **316** can be selected so as to balance overall performance.

(39) The one or more p-doped regions **316** improve performance of the emitter array **200** at higher temperatures. For example, the one or more p-doped regions **316** can increase a T_{0} parameter, which is a parameter that characterizes a threshold current I_{th} temperature dependence through the formula: $I_{th}(T) = I_{th}(0) \exp(T/T_{0})$.

(40) Notably, in some implementations, the techniques described in association with FIGS. 3A-3D can be used in any combination in order to achieve improved carrier confinement, improved carrier injection, or improved capture efficiency of the emitter array **200**. For example, the emitter array **200** may include an cavity/active region **208** including (1) one or more QWB layers **304** having an increased band gap (as described with respect to FIG. 3A), (2) one or more graded-index structures **314** (as described with respect to FIGS. 3B and 3C), and (3) one or more p-doped regions **316** (as described with respect to FIG. 3D) in order to achieve improved carrier confinement, improved carrier injection, or improved capture efficiency.

(41) In this way, performance of a dilute nitride long-wavelength emitter over temperature can be improved. For example, the emitter array **200** described herein may achieve improved carrier confinement, improved carrier injection, or improved capture efficiency by including an increased QWB band gap for a given QWB layer **304**, one or more graded-index structures **314**, or one or more p-doped regions **316** in the cavity/active region **208**. In the same way, the performance of a dilute nitride long-wavelength emitter at higher operating current could also be improved.

(42) As indicated above, FIGS. 3A-3D are provided as examples. Other examples may differ from

what is described with regard to FIGS. 3A-3D.

(43) FIG. 4 is a flowchart of an example process **400** associated with forming a dilute nitride long-wavelength emitter with improved performance over temperature, as described herein.

(44) As shown in FIG. 4, process **400** may include forming a QW layer using an epitaxial growth process, the epitaxial growth process being performed according to a first growth mode to form the QW layer (block **410**). For example, a QW layer may be formed using an epitaxial growth process, the epitaxial growth process being performed according to a first growth mode to form the QW layer, as described above.

(45) As further shown in FIG. 4, process **400** may include forming a QWB layer using the epitaxial growth process, the epitaxial growth process being performed according to a second growth mode to form the QWB layer, wherein at least one of a nitrogen flux used in the first growth mode is different from a nitrogen flux used in the second growth mode, or a gallium flux used in the first growth mode is different from a gallium flux used in the second growth mode (block **420**). For example, a QWB layer may be formed using the epitaxial growth process, the epitaxial growth process being performed according to a second growth mode to form the QWB layer, wherein at least one of a nitrogen flux used in the first growth mode is different from a nitrogen flux used in the second growth mode, or a gallium flux used in the first growth mode is different from a gallium flux used in the second growth mode, as described above.

(46) Process **400** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

(47) In a first implementation, process **400** includes forming a graded-index outer barrier layer.

(48) In a second implementation, alone or in combination with the first implementation, process **400** includes forming a p-doped region in the QWB layer.

(49) In a third implementation, alone or in combination with one or more of the first and second implementations, process **400** includes adjusting a plasma source setting associated with the epitaxial growth process, wherein a difference between the nitrogen flux used in the first growth mode and the nitrogen flux used in the second growth mode is a result of the adjusting of the plasma source setting.

(50) In a fourth implementation, in combination with the third implementation, the plasma source setting is a power setting.

(51) In a fifth implementation, in combination with one or more of the third and fourth implementations, the plasma source setting is a nitrogen gas flow setting.

(52) In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, process **400** includes switching a gallium source associated with the epitaxial growth process, wherein a difference between the gallium flux used in the first growth mode and the gallium flux used in the second growth mode is a result of the switching of the gallium source.

(53) In a seventh implementation, alone or in combination with one or more of the first through sixth implementations, process **400** includes adjusting a quantity of gallium sources used for performing the epitaxial growth process, wherein a difference between the gallium flux used in the first growth mode and the gallium flux used in the second growth mode is a result of the adjusting of the quantity of gallium sources.

(54) In an eighth implementation, alone or in combination with one or more of the first through seventh implementations, process **400** includes forming an interlayer between the QW layer and the QWB layer, where a band gap between the interlayer and the QW layer is smaller than a band gap between the QWB layer and the QW layer.

(55) In a ninth implementation, in combination with the eighth implementation, the interlayer has a thickness in a range from approximately 0.1 nanometers (nm) to 10.0 nm.

(56) In a tenth implementation, alone or in combination with one or more of the eighth and ninth implementations, the interlayer is formed by performing the epitaxial growth process according to

the first growth mode.

(57) Although FIG. 4 shows example blocks of process 400, in some implementations, process 400 includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 4. Additionally, or alternatively, two or more of the blocks of process 400 may be performed in parallel.

(58) The foregoing disclosure provides illustration and description, but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations. Furthermore, any of the implementations described herein may be combined unless the foregoing disclosure expressly provides a reason that one or more implementations may not be combined.

(59) As used herein, satisfying a threshold may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

(60) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiple of the same item.

(61) No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, or a combination of related and unrelated items), and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”). Further, spatially relative terms, such as “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the apparatus, device, and/or element in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Claims

1. A method, comprising: forming a quantum well (QW) layer using an epitaxial growth process, the epitaxial growth process being performed according to a first growth mode to form the QW layer, the QW layer comprising a first dilute nitride material; and forming a quantum well barrier (QWB) layer using the epitaxial growth process, the epitaxial growth process being performed according to a second growth mode to form the QWB layer, the QWB layer comprising a second

- dilute nitride material, wherein at least one of: a nitrogen flux used in the first growth mode is different from a nitrogen flux used in the second growth mode, or a gallium flux used in the first growth mode is different from a gallium flux used in the second growth mode.
2. The method of claim 1, further comprising forming a graded-index outer barrier layer.
 3. The method of claim 1, further comprising forming a p-doped region in the QWB layer.
 4. The method of claim 1, wherein the nitrogen flux used in the first growth mode is different from the nitrogen flux used in the second growth mode, the method further comprising adjusting a plasma source setting associated with the epitaxial growth process, wherein a difference between the nitrogen flux used in the first growth mode and the nitrogen flux used in the second growth mode is a result of the adjusting of the plasma source setting.
 5. The method of claim 4, wherein the plasma source setting is a power setting.
 6. The method of claim 4, wherein the plasma source setting is a nitrogen gas flow setting.
 7. The method of claim 1, wherein the gallium flux used in the first growth mode is different from the gallium flux used in the second growth mode, the method further comprising switching a gallium source associated with the epitaxial growth process, wherein a difference between the gallium flux used in the first growth mode and the gallium flux used in the second growth mode is a result of the switching of the gallium source.
 8. The method of claim 1, wherein the gallium flux used in the first growth mode is different from the gallium flux used in the second growth mode, the method further comprising adjusting a quantity of gallium sources used for performing the epitaxial growth process, wherein a difference between the gallium flux used in the first growth mode and the gallium flux used in the second growth mode is a result of the adjusting of the quantity of gallium sources.
 9. The method of claim 1, further comprising forming an interlayer between the QW layer and the QWB layer, where a band gap between the interlayer and the QW layer is smaller than a band gap between the QWB layer and the QW layer.
 10. The method of claim 9, wherein the interlayer has a thickness in a range from approximately 0.1 nanometers (nm) to 10.0 nm.
 11. The method of claim 9, wherein the interlayer is formed by performing the epitaxial growth process according to the first growth mode.
 12. A dilute nitride long-wavelength emitter, comprising: a cladding layer; an active region including: one or more quantum well (QW) layers, wherein the one or more QW layers comprise a first dilute nitride material; and one or more QW barrier (QWB) layers, wherein the one or more QWB layers comprise a second dilute nitride material; and a barrier layer between the active region and the cladding layer, wherein the barrier layer comprises a graded-index structure.
 13. The dilute nitride long-wavelength emitter of claim 12, wherein the graded-index structure is a graded-index separate confinement heterostructure (GRINSCH).
 14. The dilute nitride long-wavelength emitter of claim 12, wherein the graded-index structure has a linear grading.
 15. The dilute nitride long-wavelength emitter of claim 12, wherein the graded-index structure has a step grading.
 16. The dilute nitride long-wavelength emitter of claim 12, further comprising a p-doped region in at least one QWB layer of the one or more QWB layers.
 17. A dilute nitride long-wavelength emitter, comprising: an active region including: one or more quantum well (QW) layers, wherein the one or more QW layers comprise a first dilute nitride material; and one or more QW barrier (QWB) layers, wherein the one or more QWB layers comprise a second dilute nitride material, and wherein at least one QWB layer of the one or more QWB layers includes a p-doped region.
 18. The dilute nitride long-wavelength emitter of claim 17, wherein a doping level of the p-doped region is in a range from approximately $1 \times 10^{16} \text{ cm}^{-3}$ to approximately $1 \times 10^{19} \text{ cm}^{-3}$.

19. The dilute nitride long-wavelength emitter of claim 17, wherein the p-doped region has a thickness in a range from approximately 0.2 nanometers (nm) to approximately 20.0 nm.
20. The dilute nitride long-wavelength emitter of claim 17, further comprising a barrier layer comprising a graded-index structure.
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